Recreate Order Management Database

The exercises use the Order Management Database described in a separate file listed in this module. You should create the database following the instructions in the file, *SQL Single Table Exercise DB.docx* before doing the exercise. When running into problems and difficulties in doing the exercise, you should get help from your AI tool to formulate SQL query solutions and debug the queries.

Single Table Queries

Here are the exercise problems:

P-1) List item ID, name, unit cost, unit price, and discount rate of all products. Sort the result by item name in ascending order.

P-2) List discount rate, item ID, item name, unit cost, and unit price of all items. Sort the result by discount rate in descending order and item name in ascending order.

P-3) List customer ID, last name, first name, email, phone number, and address of all customers. Sort the result by last name and address in ascending order.

P-4) For all order lines in the database, list order number, order line ID, item ID, quantity, and unit sales price. Sort the result by order number and order line ID in ascending order.

P-5) For every item with a discount rate between 3% and 5%, list discount rate, item ID, item name, unit cost, and unit price. Sort the result by the discount rate in descending order and the item name in ascending order.

P-6) For customers whose street name contains "Peachtree", list customer ID, last name, first name, and address. Sort the result by the last name in ascending order.

P-7) For customers who live in either Decatur or Atlanta, list customer ID, last name, first name, and address. Sort the result by the last name in ascending order.

P-8) For all items sold, list those with item ID `P201`, `P203`, `P204`, and `P206`. Sort the result by item ID in ascending order and quantity in descending order. (*Note that some items may have no sales record depending on the state of the database state when the query is run. Your SQL statement must guarantee to produce correct answers in any given database state.*)

P-9) For order lines whose order number is `O401`, `O403`, `O407` or `O408`, list order number, order line ID, item ID, quantity, unit sales price, and order line total which is the quantity multiplied by the unit sales price. Give a descriptive name for the computed field. Sort the result by order number, order line ID, and item ID in ascending order.

P-10) List item ID, item name, unit cost, unit price, discount rate, and discounted unit price which is the unit price minus the discount. Give a descriptive name for the computed field. Sort the result by item ID.

P-11) For order lines whose line total is equal to or greater than 100, list order number, order line ID, item ID, quantity, unit sales price, and order line total. Give a descriptive name for the computed field. Sort the result by order line total, order number in descending order, and order line ID and item ID in ascending order.

P-12) For those items whose unit cost is equal to or greater than $30, list item ID, item name, unit cost, unit price, discount rate, and discounted unit price that is unit price minus the discount. Give a descriptive name for the computed field. Sort the result by item ID.

P-13) For items whose discounted unit price is greater than $50, list item ID, item name, unit cost, unit price, discount rate, and discounted unit price. Give a descriptive name for the computed field. Sort the result by discounted unit price in descending order and item ID in ascending order.

P-14) For items for which the discount dollar amount is greater than $2.5, list item ID, item name, unit cost, unit price, discount rate, and discounted unit price. Give a descriptive name for the computed field. Sort the result by discounted unit price in descending order and item ID in ascending order.

P-15) The user wants a report about orders. The report should show the order number and the total sales amount of the order. Give a descriptive name to the computed data in the result. Sort the result by order total and order number in ascending order.

P-16) The user wanted a report to show the largest sales amount, the smallest sales amount, and the average sales amount of all orders. Give descriptive names to all computed data in the result.